



United International University
QUEST FOR EXCELLENCE

PROJECT REPORT

VLSI Design

Course Code: CSE 4327

Course Section: A

Project Title: *Design, Verification, and Synthesis of a Single-Round DES Datapath*

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1. Motivation and Scope

This project implements one complete DES round as a combinational datapath. The objective was to apply RTL design, simulation, and standard-cell synthesis to a cryptographic hardware block.

The **Data Encryption Standard**, or DES, was conceived in the early 1970s and can arguably be considered the first modern encryption algorithm. It was the most popular block cipher in the 1980s and 1990s. Even though DES in its basic form is nowadays not secure because of the small key space, its variant 3DES or triple DES is still in use in the 2020s. The design principles of DES have inspired many current ciphers and, hence, studying it helps us to understand many other symmetric algorithms. (Ref: *Understanding Cryptography* by Christoff Paar)

As DES is one of the most foundational block cipher algorithms in Cryptography, understanding the basics is very important for us to navigate the world of modern cybersecurity. Due to these ciphers being widely used, often specialized **ASICs** are used to compute them. Our primary motivation for this project was to understand how we can convert the abstract mathematical and algorithmic concepts into hardware implementations.

DES Encryption Process

1. DES accepts a **64-bit plaintext block** and a **64-bit key**. Only 56 key bits are used for encryption; the remaining 8 bits are parity bits.
2. The plaintext first passes through the **Initial Permutation (IP)**, which rearranges the 64 input bits according to a fixed table.
3. The permuted data is divided into two 32-bit halves: the left half (L_0) and the right half (R_0).
4. The key schedule generates **sixteen 48-bit round keys**, one for each of the 16 DES rounds.
5. In each round, the right half is expanded from 32 bits to 48 bits using the expansion permutation.
6. The expanded right half is XORed with that round's 48-bit key.
7. The 48-bit result is divided into eight 6-bit groups. Each group passes through one DES S-box, producing eight 4-bit outputs. Together, the S-boxes convert 48 bits back into 32 bits.
8. The 32-bit S-box output passes through the fixed **P permutation**.
9. The permuted result is XORed with the previous left half. The previous right half becomes the new left half. This completes one Feistel round.
10. DES repeats the same round operation **16 times**, using a different round key every time.
11. After the final round, the two 32-bit halves are combined and passed through the **Final Permutation**, which is the inverse of the initial permutation.
12. The resulting 64-bit value is the ciphertext.

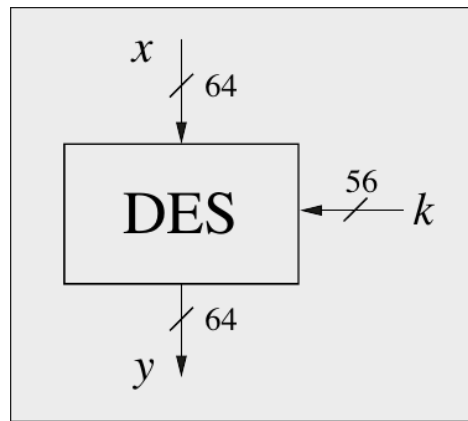


Figure 1. Block diagram of DES

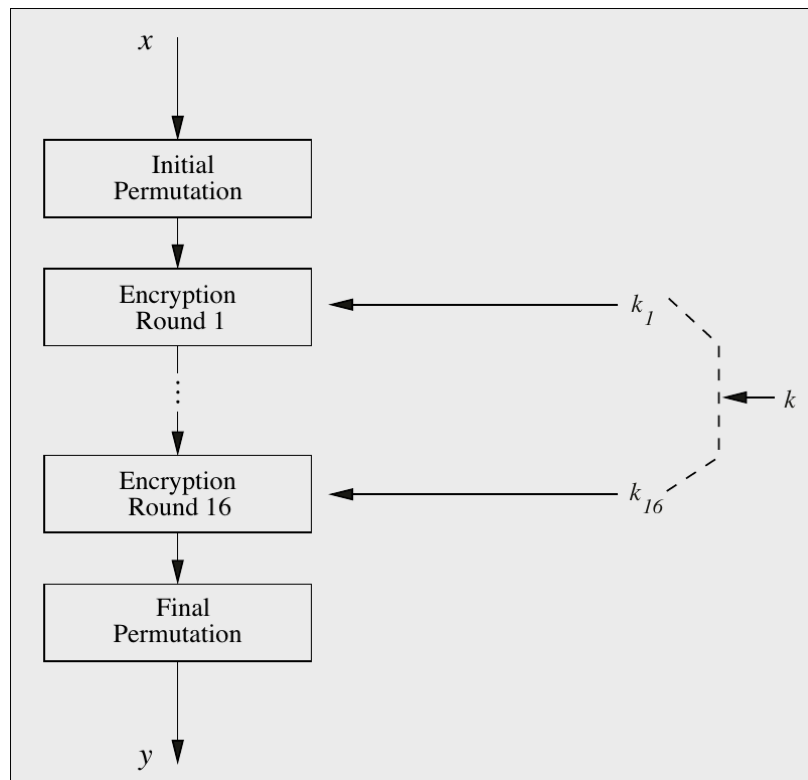
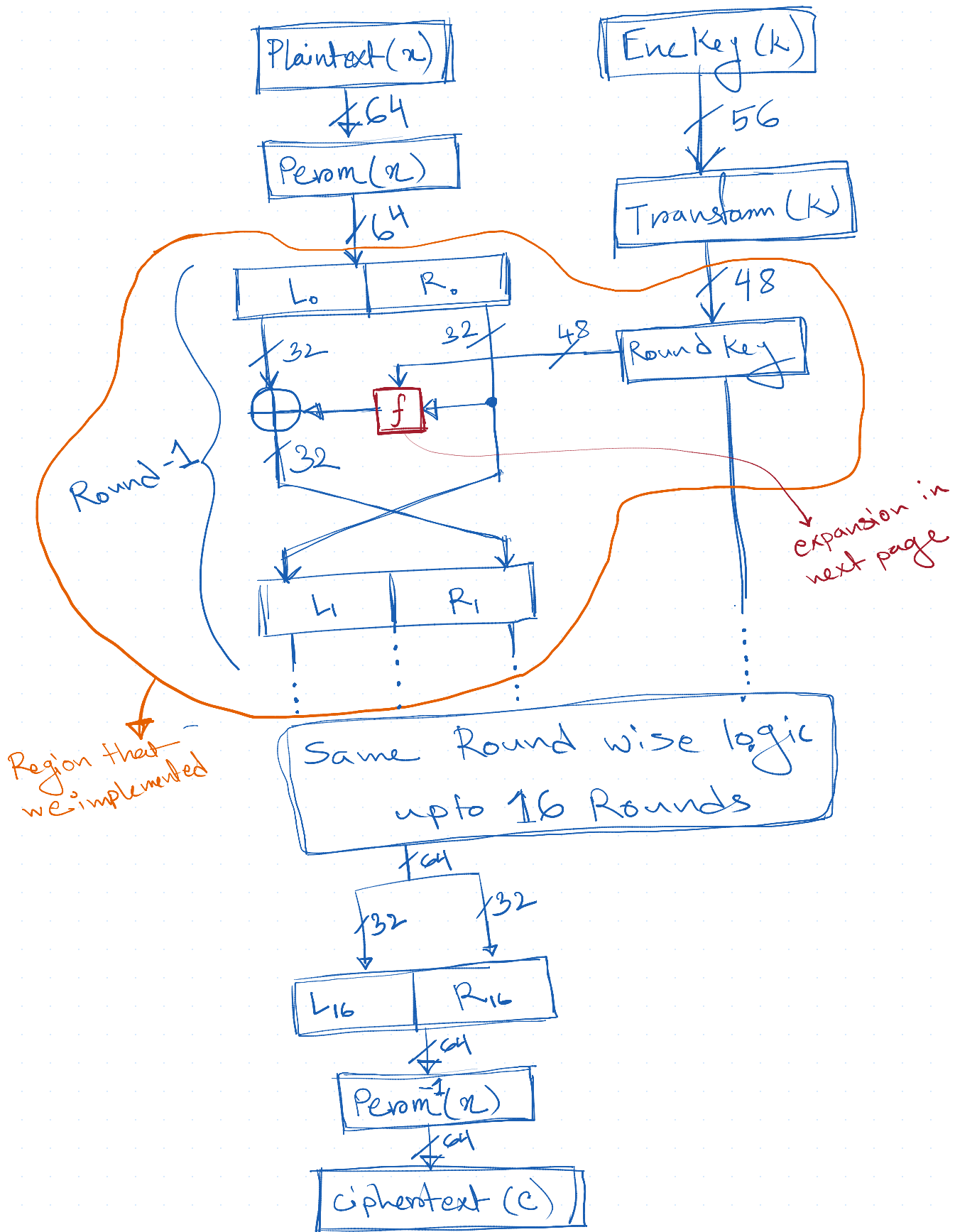


Figure 2. Round diagram of DES

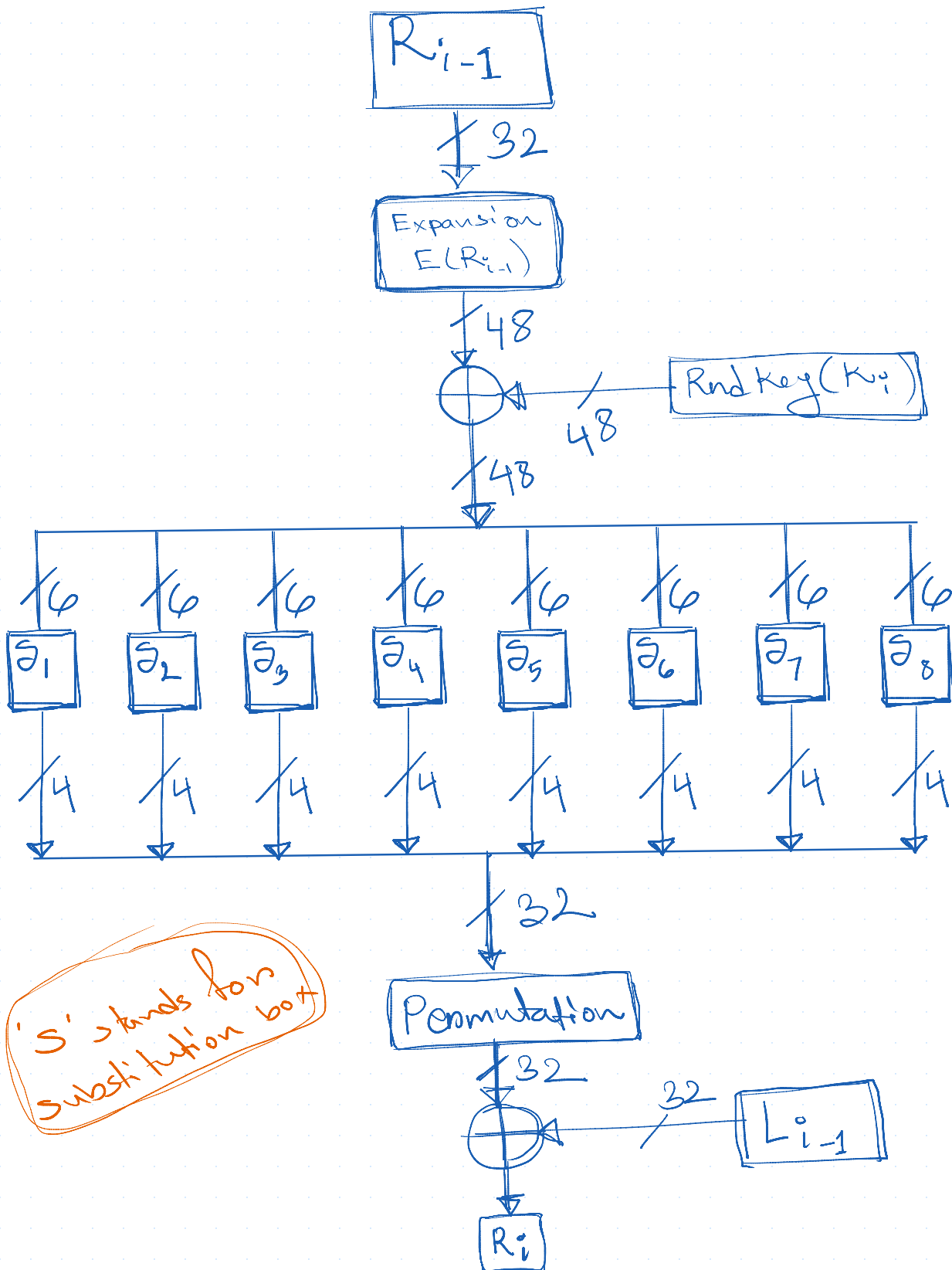
Scope:

For this project, only the logic in Steps **5 to 9** is implemented for one round. The 64-bit state and 48-bit round key are given directly as inputs. Detailed Diagram is shown below:

Core DES Algorithm



The "f" function expanded



2. Methodology

1. First, a **C implementation** of one DES round was developed. It performs expansion, round-key XOR, all eight S-box substitutions, P permutation, and the Feistel operation. Its output was used as the reference for the Verilog testbench. The C files are stored in:

codes/C_implementation/

2. The same datapath was then implemented as a combinational Verilog module. The 64-bit input is divided into two 32-bit halves. The right half is expanded to 48 bits, XORed with the supplied 48-bit round key, processed through eight S-boxes, and passed through the P permutation. The final Feistel operation produces the 64-bit output. The RTL and testbench are stored in:

codes/Verilog/des_round.v

codes/Verilog/tb_des_round.v

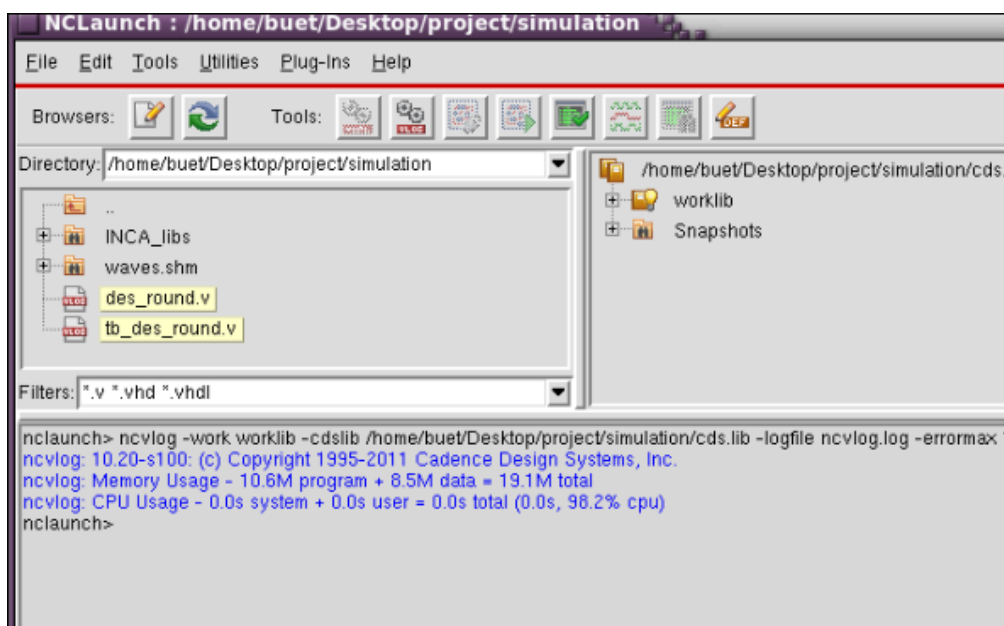


Figure 3. Successful Compilation

3. The design was simulated in the Cadence environment using six directed test cases. Each testbench output was compared with the expected C-model result. The simulation files, logs, and waveform database are stored in:

artifacts/VM_outputs/simulation/

4. Finally, the verified RTL was synthesized using Cadence RTL Compiler and the slow.lib standard-cell library. A 10 ns virtual clock with input and output delay constraints was used because the design is combinational. The constraints, library, RTL, and synthesis outputs are stored in:

artifacts/VM_outputs/constraints/

artifacts/VM_outputs/lib/

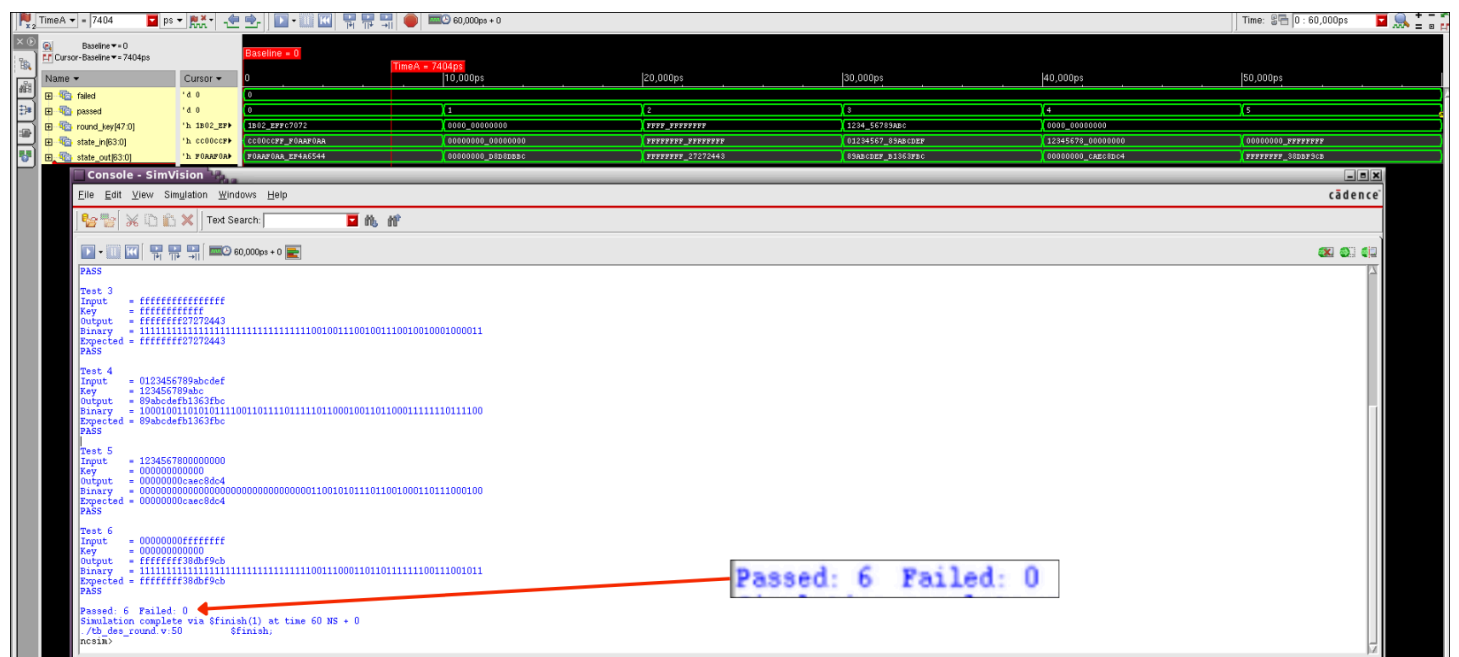
artifacts/VM_outputs/rtl/

artifacts/VM_outputs/synthesis/

- report/images/

The testbench applies six directed test cases and compares each hardware output against the expected C-reference output.

Test	Input State	Round Key	Expected Output	Result
1	CC00CCFFF0AAF0AA	1B02EFFC7072	F0AAF0AAEF4A6544	PASS
2	0000000000000000	000000000000	00000000D8D8DBBC	PASS
3	FFFFFFFFFFFFFFFF	FFFFFFFFFFFFFF	FFFFFFFFF27272443	PASS
4	0123456789ABCDEF	123456789ABC	89ABCDEFB1363FBC	PASS
5	1234567800000000	000000000000	00000000CAEC8DC4	PASS
6	00000000FFFFFFFF	000000000000	FFFFFFFFF38DBF9CB	PASS



The final testbench result was:

4. Synthesis Results

Cadence RTL Compiler was used to synthesize the Verilog RTL using the slow.lib standard-cell library. The synthesis generated a mapped gate-level netlist, timing constraints, area report, timing report, power report, and schematic.

```
rc:/> report area
```

```
=====
```

```
Generated by:      Encounter(R) RTL Compiler v12.10-s012_1
```

```
Generated on:      Oct 05 2026  06:34:38 am
```

```
Module:            des_round
```

```
Technology library: gpdK045bc
```

```
Operating conditions: slow (balanced_tree)
```

```
Wireload mode:     enclosed
```

```
Area mode:         timing library
```

```
=====
```

Instance	Cells	Cell Area	Net Area	Total Area	Wireload
des_round	838	0	0	0	<none> (D)

```
(D) = wireload is default in technology library
```

```
rc:/> █
```

Figure 5. RTL Compiler area report.

```
rc:/> report timing
```

```
Warning : Possible timing problems have been detected in this design. [TIM-11]
```

```
          : The design is 'des_round'.
```

```
=====
```

```
Generated by:      Encounter(R) RTL Compiler v12.10-s012_1
```

```
Generated on:      Oct 05 2026  06:36:02 am
```

```
Module:            des_round
```

```
Technology library: gpdK045bc
```

```
Operating conditions: slow (balanced_tree)
```

```
Wireload mode:     enclosed
```

```
Area mode:         timing library
```

```
=====
```

Pin	Type	Fanout	Load (fF)	Slew (ps)	Delay (ps)	Arrival (ps)	
(clock VCLK)	launch					0	R
(des_round.g_line_4_74_1)	ext delay			+1000		1000	F
round_key[37]	in port	1	1.7	0	+0	1000	F
g16698/S0					+0	1000	
g16698/Y	MXI2X1	4	18.9	641	+370	1370	F
g16280/B					+0	1370	
g16280/Y	NAND2X8	8	33.0	194	+224	1594	R
g15769/A0					+0	1594	
g15769/Y	OAI222X4	2	8.2	228	+245	1840	F
g15704/B					+0	1840	
g15704/Y	NOR2X4	2	3.0	80	+101	1940	R
g16706/AN					+0	1940	
g16706/Y	NOR4BX4	1	2.9	147	+196	2136	R
g15624/D					+0	2136	
g15624/Y	NAND4BBX4	1	1.5	142	+115	2251	F
g16627/B					+0	2251	
g16627/Y	XNOR2X4	1	0.0	56	+208	2459	R
state_out[19]	out port				+0	2459	R
(des_round.g_line_7_266_1)	ext delay			+1000		3459	R
(clock VCLK)	capture					10000	R

```
-----
```

```
Cost Group : 'VCLK' (path_group 'VCLK')
```

```
Timing slack : 6541ps
```

```
Start-point : round_key[37]
```

```
End-point : state_out[19]
```

Figure 6. RTL Compiler timing report.

```
rc:/> report power
```

```
=====
```

```
Generated by:      Encounter(R) RTL Compiler v12.10-s012_1
```

```
Generated on:      Oct 05 2026  06:35:25 am
```

```
Module:           des_round
```

```
Technology library: gpdk045bc
```

```
Operating conditions: slow (balanced_tree)
```

```
Wireload mode:    enclosed
```

```
Area mode:        timing library
```

```
=====
```

		Leakage	Dynamic	Total
Instance	Cells	Power(nW)	Power(nW)	Power(nW)
des_round	838	214.738	536372.182	536586.919

Figure 7. RTL Compiler power report.

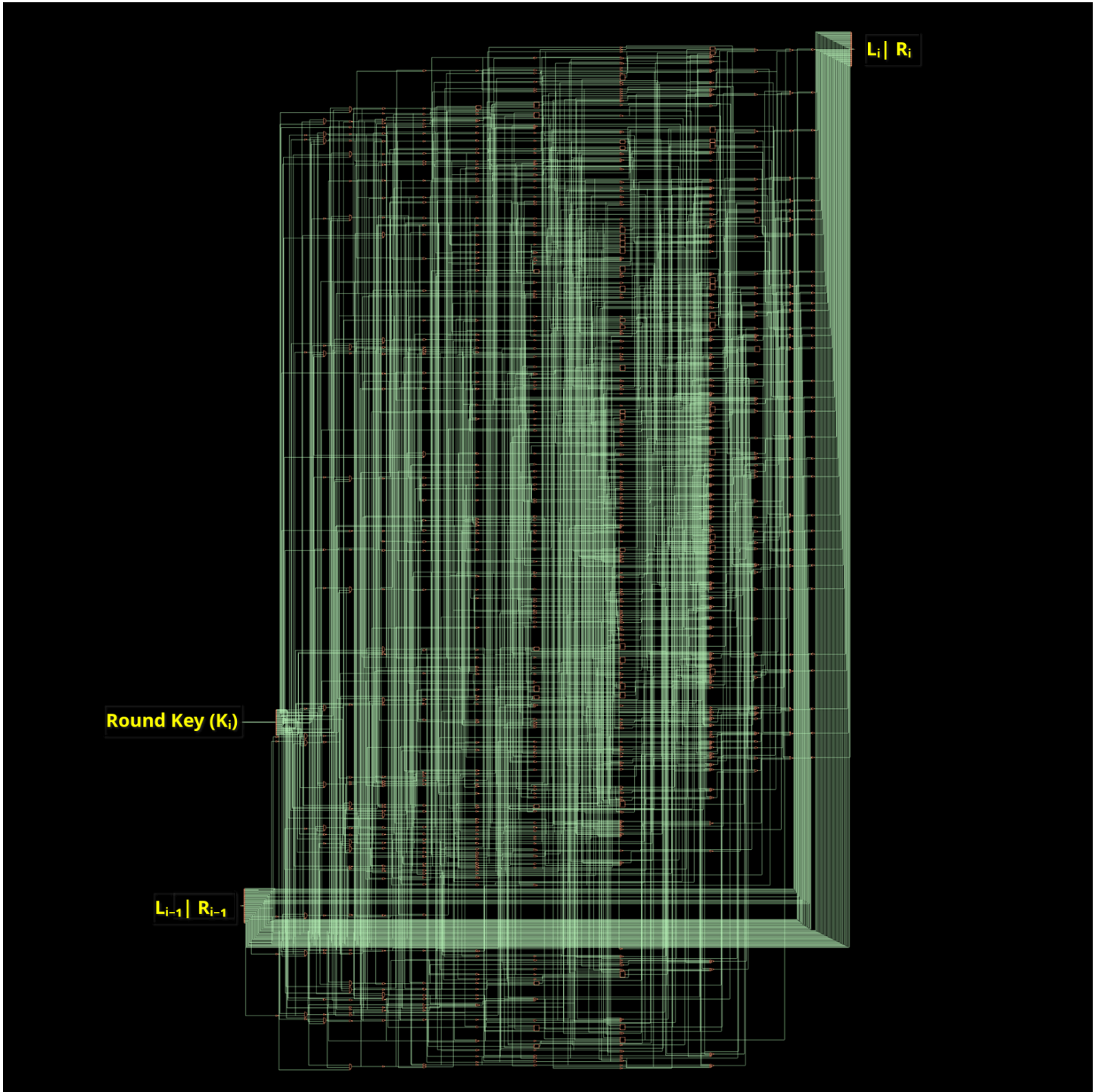


Figure 8. Synthesized standard-cell schematic of the single-round DES datapath.

PS: A more high resolution schematic is available in the *artifacts/schematics* directory

Metric	Result
Total cell area	0
Number of cells	838
Worst slack	6.541 ns
Total power	536586.919 nW

The design has no sequential cells because it is fully combinational. The S-boxes are expected to contribute most of the logic complexity, while the expansion and P permutation are mainly fixed wiring operations.

5. Conclusion and Files

The project successfully implemented and verified a single-round DES datapath using C and Verilog, followed by standard-cell synthesis using Cadence RTL Compiler. The design includes the complete internal DES round function—expansion, round-key mixing, all eight S-boxes, P permutation, and Feistel output generation. The Verilog implementation passed all six directed test cases against the expected outputs generated by the C reference model. RTL Compiler also generated the required synthesis reports, mapped netlist, timing constraints, power results, and gate-level schematic. Although the design does not include the DES key schedule, initial/final permutations, or the remaining fifteen rounds, it demonstrates the complete flow from algorithmic reference modeling to RTL verification and standard-cell synthesis.

The project files and results are organized as follows:

- C reference implementation, execution script, and output: *codes/C_implementation/*
- Verilog RTL module and self-checking testbench: *codes/Verilog/*
- Timing constraints: *artifacts/VM_outputs/constraints/*
- Standard-cell library: *artifacts/VM_outputs/lib/*
- RTL source used for synthesis: *artifacts/VM_outputs/rtl/*
- Cadence simulation files and logs: *artifacts/VM_outputs/simulation/*
- RTL Compiler synthesis outputs: *artifacts/VM_outputs/synthesis/*
- Area report: *artifacts/VM_outputs/synthesis/area.rpt*
- Gate report: *artifacts/VM_outputs/synthesis/gates.rpt*
- Timing report: *artifacts/VM_outputs/synthesis/timing.rpt*
- Power report: *artifacts/VM_outputs/synthesis/power.rpt*
- Mapped netlist: *artifacts/VM_outputs/synthesis/des_round_netlist.v*
- Exported constraints: *artifacts/VM_outputs/synthesis/des_round.sdc*
- Synthesized schematic: *artifacts/schematic.png* and *artifacts/schematic.pdf*
- Report screenshots: *report/images/*
- DES block diagram: *report/image.png*